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Title: Analysis of the expected L_2 error of an over-parametrized deep neural network estimate learned by gradient descent without regularization.

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Primary classification:

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Review text:

This paper studies an over-parametrized deep neural network (DNN) estimator in nonparametric regression. The main novelty is that the estimator is trained **without explicit regularization**, using only gradient descent on the empirical L_2 loss. Under suitable choices of network architecture, initialization, step size, and number of gradient steps, the authors prove universal consistency and rates of convergence, extending also to interaction models that circumvent the curse of dimensionality.

Let $(X, Y) \in \mathbb{R}^d \times \mathbb{R}$ with $\mathbb{E}Y^2 < \infty$ and regression function $m(x) = \mathbb{E}[Y \mid X = x]$. Given i.i.d. data $\{(X_i, Y_i)\}_{i=1}^n$, the network output is a linear combination of K_n fully connected networks with L layers, r neurons per layer, and logistic squasher $\sigma(x) = 1/(1 + e^{-x})$:

$$f_{\mathbf{w}}(x) = \sum_{k=1}^{K_n} w_{1,1,k}^{(L)} f_{k,1}^{(L)}(x),$$

with $f_{k,i}^{(l)}$ defined recursively via

$$f_{k,i}^{(l)}(x) = \sigma \left(\sum_{j=1}^r w_{k,i,j}^{(l-1)} f_{k,j}^{(l-1)}(x) + w_{k,i,0}^{(l-1)} \right).$$

Weights are learned by gradient descent on the empirical risk

$$F_n(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}}(X_i) - Y_i|^2,$$

with step size $\lambda_n = 1/t_n$ and t_n steps. Initialization sets outer weights to zero and inner weights random in intervals scaling with $\log n$ and n^τ . The final estimate is truncated: $m_n(x) = T_{\beta_n} f_{\mathbf{w}(t_n)}(x)$ with $\beta_n = c_1 \log n$.

Universal consistency (Theorem 3.1). Assume $\text{supp}(X)$ bounded, $\mathbb{E}Y^2 < \infty$, and for some $\kappa > 0$,

$$\frac{K_n}{n^\kappa} \rightarrow 0, \quad \frac{K_n}{n^{r+2}} \rightarrow \infty,$$

and $t_n \geq c_2 L_n$ with $L_n \geq K_n^{3/2} (\log n)^{6L+5}$. Then

$$\mathbb{E} \int |m_n(x) - m(x)|^2 d\mathbb{P}_X(x) \rightarrow 0 \quad (n \rightarrow \infty).$$

The condition $K_n \gg n^{r+2}$ formalizes over-parametrization, yet consistency holds.

Rate of convergence (Theorem 3.2). Let m be (p, C) -Hlder smooth with $1/2 \leq p \leq 1$, $\text{supp}(X) \subseteq [0, 1]^d$, and $\mathbb{E}e^{c_3 Y^2} < \infty$. Choose

$$K_n = n^{6d+r+2}, \quad \tau = \frac{1}{1+d}, \quad t_n \approx K_n^{3/2} (\log n)^{6L+2}.$$

Then for any $\epsilon > 0$,

$$\mathbb{E} \int |m_n(x) - m(x)|^2 d\mathbb{P}_X(x) \leq c_5 n^{-\frac{1}{1+d} + \epsilon}.$$

The rate $n^{-1/(1+d)}$ is optimal for $p = 1/2$; for $p > 1/2$ it does not improve due to approximation error bounds.

Interaction models (Theorem 3.3). If

$$m(x) = \sum_{|I|=d^*} m_I(x_I), \quad x_I = (x^{(j_1)}, \dots, x^{(j_{d^*})}),$$

with (p, C) -smooth components, then

$$\mathbb{E} \int |m_n(x) - m(x)|^2 d\mathbb{P}_X(x) \leq c_7 n^{-\frac{1}{1+d^*} + \epsilon},$$

independent of the original dimension d , showing dimension reduction.

A key technical tool is **Lemma 5.1**, which bounds the optimization error under convexity in part of the parameters. For weight vectors (u, v) with $u \mapsto F(u, v)$ convex, step size $\lambda = 1/t_n$, one obtains

$$F(u_{t_n}, v_{t_n}) \leq F(u^*, v_0) + D_n \|u^*\| \sqrt{2F(u_0, v_0)} + \frac{\|u^* - u_0\|^2}{2} + \frac{F(u_0, v_0)}{t_n}.$$

Lemma 5.7 provides an approximation bound: for $l \approx \frac{1}{1+d} \log_2 n$ and $\delta = n^{-1/(1+d)}$,

$$\int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} f_{\tilde{w},k,1}^{(L)}(x) - m(x) \right|^2 d\mathbb{P}_X(x) \leq c \left(n^{-\frac{1}{1+d}} + n^{-\frac{2p}{1+d}} \right).$$

Complexity is controlled via metric entropy (Lemma 5.4), enabling empirical process bounds. The removal of the regularization term is a significant conceptual contribution, and the simulation study illustrates practical relevance.